

E-commerce Sales Analysis project.

SIC – AI702

Agenda

- Team Members
- Overview
- Problem Statment
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- E.D.A
- Analyzing e-commerce sales using Power BI

Team Members



 [Mohamed Elzagh](#)



 [mariam sherif](#)



 [Ahmed Hany](#)



Overview

Overview

This project focuses on conducting data analysis on E-commers data and finding key insights on the business using visualization tools and data manipulation tools to extract question to help the business in solving problems.

The ultimate goal is to transform raw data into an intuitive and interactive dashboard that empowers decision-makers.



Problem statement

Problem statement

- Which of our products are the true top-sellers?
- Which geographical regions are driving the most revenue?
- How do our sales fluctuate over time, and are there seasonal trends we can capitalize on?
- Which month has the most profit?
- Which customers has the most purchases?
- What is the most used Payment Method?
- What is the profit margin percentage?

Data preparation

Data Overview

Two primary datasets were used for this analysis

- **Details.csv:**

- contains transactions info

details_df

	Order ID	Amount	Profit	Quantity	Category	Sub-Category	PaymentMode
0	B-25681	1096	658	7	Electronics	Electronic Games	COD
1	B-26055	5729	64	14	Furniture	Chairs	EMI
2	B-25955	2927	146	8	Furniture	Bookcases	EMI
3	B-26093	2847	712	8	Electronics	Printers	Credit Card
4	B-25602	2617	1151	4	Electronics	Phones	Credit Card
...	...	...	...	...	...	...	...
1495	B-25700	7	-3	2	Clothing	Hankerchief	COD
1496	B-25757	3151	-35	7	Clothing	Trousers	EMI
1497	B-25973	4141	1698	13	Electronics	Printers	COD
1498	B-25698	7	-2	1	Clothing	Hankerchief	COD
1499	B-25993	4363	305	5	Furniture	Tables	EMI

1500 rows × 7 columns

- **Orders.csv:**

- contains order info

orders_df

	Order ID	Order Date	CustomerName	State	City
0	B-26055	10-03-2018	Harivansh	Uttar Pradesh	Mathura
1	B-25993	03-02-2018	Madhav	Delhi	Delhi
2	B-25973	24-01-2018	Madan Mohan	Uttar Pradesh	Mathura
3	B-25923	27-12-2018	Gopal	Maharashtra	Mumbai
4	B-25757	21-08-2018	Vishakha	Madhya Pradesh	Indore
...	...	...	...	...	...
495	B-25742	03-08-2018	Ashwin	Goa	Goa
496	B-26088	26-03-2018	Bhavna	Sikkim	Gangtok
497	B-25707	01-07-2018	Shivani	Maharashtra	Mumbai
498	B-25758	22-08-2018	Shubham	Himachal Pradesh	Simla
499	B-26095	28-03-2018	Monisha	Rajasthan	Jaipur

500 rows × 5 columns

Data preprocessing

Checked the two csv's

No missing data found

No duplicate records

Data format standrized

Merged Details.csv & Orders.csv into cleaned_online_sales_data.csv by the unique Order ID in both cvs's

```
--- Merged DataFrame Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Order ID        1500 non-null   object
1   Amount          1500 non-null   int64
2   Profit          1500 non-null   int64
3   Quantity        1500 non-null   int64
4   Category        1500 non-null   object
5   Sub-Category    1500 non-null   object
6   PaymentMode     1500 non-null   object
7   Order Date      1500 non-null   datetime64[ns]
8   CustomerName    1500 non-null   object
9   State          1500 non-null   object
10  City            1500 non-null   object
dtypes: datetime64[ns](1), int64(3), object(7)
memory usage: 129.0+ KB

Missing values in Merged DataFrame:
Order ID    0
Amount      0
Profit      0
Quantity    0
Category    0
```



E.D.A

Statistical insights

Average order amount

₹291.85

Average Profit

₹24.64

Average Quantity

3.74

Profit Range

₹-1981 to ₹1864

Distribution Insights:

- The data shows a right-skewed distribution for Amount and Profit, with a significant number of transactions generating negative profit. This suggests potential pricing or cost issues in certain product categories that need further investigation.

Key Insights:

- High variability in profit
- Significant number of transactions with negative profit
- Median profit (₹8) much lower than mean profit (₹24.64), indicating skewed distribution
- Quantity per order ranges from 1 to 14 items

Correlation

We found 2 weak positive correlation and 1 with nearly no correlation between:

- Amount / Profit: 0.309
- Amount / Quantity: 0.352
- profit / Quantity: 0.06

```
# 9. Correlation Matrix for numerical variables
print("\n--- Correlation Matrix ---")
print(df[["Amount", "Profit", "Quantity"]].corr())
```

```
--- Correlation Matrix ---
          Amount    Profit  Quantity
Amount    1.000000  0.309242  0.352486
Profit    0.309242  1.000000  0.063096
Quantity  0.352486  0.063096  1.000000
```

Amount Vs Profit

We found a correlation between Amount and Profit

Amount vs. Profit Correlation ($r = 0.309$)

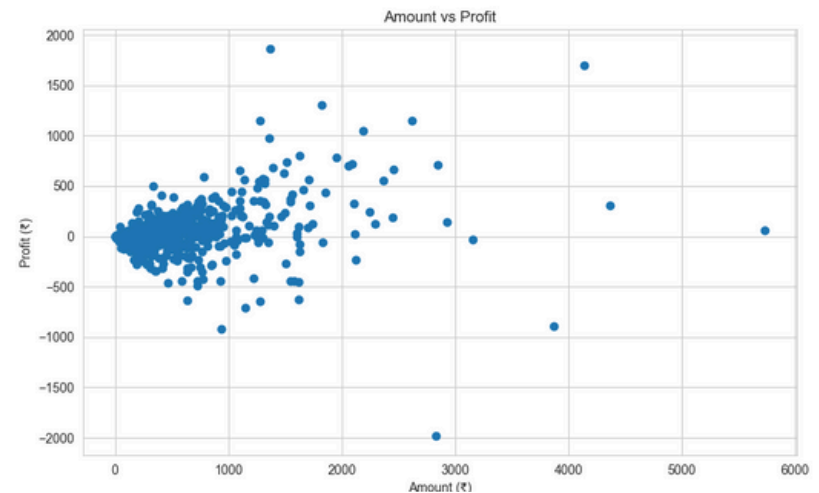
Its a weak positive relationship; as the amount increase the profit slightly increase.

Why not stronger?

- High sales often come with discounts or promotions
- Cost differences across products reduce profit margins
- expensive items are not always the most profitable

Key Insight:

Profit is driven more by margins and cost control than by sales amount alone



Amount Vs Quantity

We found a correlation between the Amount and Quantity

Orders Vs. Quantity Correlation ($r = 0.352$)

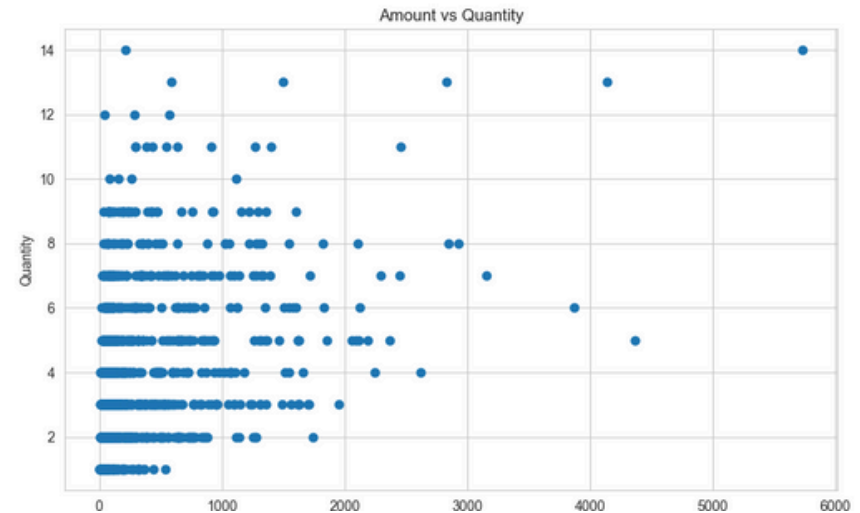
while Order sales value = item quantity $\times$ price, and adding items usually increases sales value, but not always linearly.

Why?

- Wide range of low-cost vs. high-cost items
- A single high-cost item can exceed the sales value of many low-cost items

Key insight:

Sales value is shaped not just by how many items are sold, but by which items are sold



Analyzing e-commerce sales using Power BI

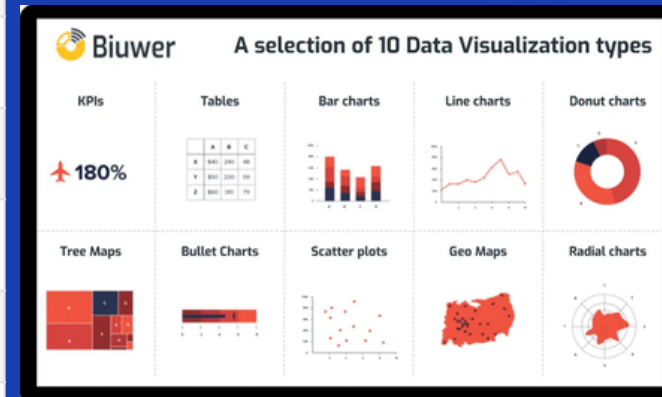
The importance of sales data analysis

- Improve understanding of customer behavior and purchasing trends
- Identify the most profitable and least performing products
- Sales analysis by geographic region
- Understanding payment method preferences and their impact on sales
- Making data-driven business decisions



Sales data structure

Field name	Description
Order ID	Unique order identification number
Amount	Order value in local currency
Profit	Profit from order
Quantity	Number of units sold
Category	Product Category (Electronics, Clothing, Furniture)
Sub-Category	Product subcategory
PaymentMode	Payment method used



Key Insights

Organizing data in this way allows analysis of the relationship between product categories, payment methods, and profitability, helping to determine which products have the highest profit margins and which payment methods are most popular with customers.

Order Dashboard



Key points

- **Monthly Trends:** The dashboard shows profit and amount trends by month, helping to identify seasonal patterns.
- **Best-selling products:** Displays the best-selling subcategories, helping to improve inventory strategies.
- **Sales breakdown by category:** Analysis of sales ratios between major categories (clothing, electronics, furniture).
- **Monthly Details:** A table showing the quantity and amount details by category for each month



Insights gained

The data shows that the profit margin (8.44%) is relatively low, indicating a need to review pricing strategies. There is also significant variation in profitability across categories, with electronics achieving higher profit margins than clothing and furniture

Customer Dashboard

Customer Behavior Summary

The analyzed client made a total of \$437,771, with a profit of \$36,907, and the preferred payment method was cash on delivery (COD).



Key points

- Customer Contribution: Analyze each customer's contribution to total sales.
- Monthly Profit Trends: Track your most profitable periods.
- Payment Methods: Analyze sales distribution by payment method.



Insights gained

- Customers' preference for cash on delivery (COD) and credit cards underscores the importance of providing diverse payment options. Fluctuating monthly profits also call for developing seasonal marketing strategies to improve sales during less active periods

Online Sales Dashboard

Main features

- Multiple filters: filter by category, subcategory, and state.
- Geographic Analysis: Sales Distribution Map by State.



Key points

- Customer Contribution: Analyze each customer's contribution to total sales.
- Monthly Profit Trends: Track your most profitable periods.
- Payment Methods: Analyze sales distribution by payment method.



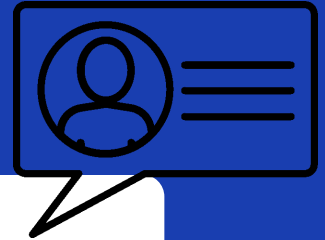
Insights gained

- Data shows that the furniture category represents the largest portion of sales (55%), indicating the importance of focusing on this category in marketing strategies. Geographic analysis also shows that sales are concentrated in specific regions, presenting an opportunity for expansion into less active areas.

Conclusion and recommendations

Main results

- The furniture category represents the largest percentage of sales, followed by electronics and then clothing.
- Cash on Delivery (COD) is the preferred method for customers.
- There is a variation in profitability between different categories, with electronics generating a higher profit margin.



Recommendations

Improve inventory strategy

- Focus on higher-margin products and improve inventory management for less profitable products.

Developing marketing strategies

- Target top-performing geographies and tailor marketing campaigns to lower-selling categories.

Diversify payment methods

- Encourage the use of electronic payment methods by offering discounts or additional benefits.

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